

PROJECT BRIEF

NEWCO MULTI-CHANNEL MARKETING CAMPAIGN PERFORMANCE AND STRATEGY OPTIMIZATION

PROJECT BACKGROUND

NewCo, a major big-box retailer, recently executed a multi-channel digital advertising campaign aimed at driving business growth and refining its long-term marketing strategy. The campaign evaluated different combinations of messaging strategies (sales-focused vs. brand-focused) and marketing channels (such as Email, Instagram, and web banners).

Partnering with BCG X, NewCo seeks to analyze the post-campaign performance data to evaluate effectiveness across formats, messaging types, and target customer segments. As part of the cross-functional BCG X team blending technical data science with management consulting, this project translates raw marketing metrics into strategic guidance for C-suite decision-makers.

1. BUSINESS PROBLEM

NewCo invested significant budget across multiple digital marketing channels and messaging strategies without an established baseline for channel-message efficiency or new customer acquisition drivers. With the campaign concluded, leadership lacks clarity on which specific combinations created true incremental value versus those that resulted in inefficient spend. Without data-driven insights into engagement drivers and channel ROI, NewCo risks misallocating marketing resources in upcoming quarters.

2. PROJECT OBJECTIVES

- **Performance Evaluation:** Quantify and compare the effectiveness of different marketing messages (sales vs. brand) across digital channels and customer segments.
- **New Customer Acquisition Analysis:** Identify the key engagement drivers, messaging themes, and channels that successfully acquired and converted new customers.
- **Strategic Budget Allocation:** Formulate data-backed recommendations on where to focus marketing investments next quarter to optimize return on ad spend (ROAS) and long-term customer value.
- **Executive Presentation:** Translate complex statistical and performance metrics into clean, high-impact visualizations and actionable strategic narratives for non-technical stakeholders.

3. SCOPE OF WORK

PROJECT OVERVIEW

Data Analyst: BCG X Data Science Team

Client/Sponsor: NewCo Marketing and Executive Leadership

Purpose

Analyze post-campaign performance data across marketing channels, messaging strategies, and customer segments to evaluate marketing spend efficiency, identify key drivers of new customer acquisition, and provide actionable recommendations for next quarter's budget allocation.

SCOPE / MAJOR PROJECT ACTIVITIES

Activity	Description
Data Cleaning and Preprocessing	Audit raw campaign logs, handle missing/anomalous records, standardize schemas, and construct key analytical metrics (CPA, CTR, Conversion Rate).
Exploratory Performance Analysis	Conduct comparative analysis evaluating performance across channel-messaging pairs, creative formats, and customer demographics.
Customer Segment and Driver Analysis	Isolate and evaluate drivers of new customer acquisition versus existing customer re-engagement.
Dashboard and Asset Creation	Build interactive dashboards and visual charts illustrating performance KPIs and budget allocation models.
Executive Synthesis and Reporting	Develop strategic narrative, formal documentation, and leadership presentation deck summarizing core recommendations.

OUT OF SCOPE

This project does not include:

- Analysis of offline, in-store, or non-digital campaign channels outside the provided dataset.
- Historical multi-year campaign trend analysis (analysis is bounded strictly to the recent campaign window).
- Real-time ad server integration, API pipelines, or automated deployment of live bidding models.

DELIVERABLES

Deliverable	Description / Details
Project Brief and Data Dictionary	Comprehensive documentation defining problem statement, dataset schema, and data pipeline preparation.
Excel Workbench and Models	Structured Excel workbook containing cleaned raw dataset, Pivot Tables, summary matrices, and slicers.
Analysis Report	Formal document detailing findings on channel ROI, messaging impact, and customer acquisition drivers.
Interactive Dashboard	Visual interface showing high-level KPIs, channel performance drills, and customer segmentation filters.
Executive Presentation Deck	Slide deck tailored for executive decision-makers featuring strategic recommendations for Q4 spend.

SCHEDULE OVERVIEW / MAJOR MILESTONES

Milestone	Expected Completion Date	Description / Details
Project Framing and Data Prep	Phase 1	Finalize Project Brief, clean dataset, and document Data Dictionary.
Excel pivot Analysis	Phase 2	Build summary pivot tables (Channel, Campaign, Channel x Campaign) and apply Slicers to isolate New customer metrics.

Excel Dashboard and Reporting	Phase 3	<i>Construct interactive Excel charts/dashboards and draft core Analysis Report summarizing engagement drivers.</i>
Executive Deck and Final Delivery	Phase 4	<i>Finalize slide deck and deliver end-to-end documentation suite.</i>

Estimated date for completion: End of Simulation Phase

4. KEY BUSINESS QUESTIONS

- I. **Value and ROI:** Which campaign and channel combination delivered the highest total value and conversion effectiveness?
- II. **Customer Engagement:** What messaging types (sales-focused vs. brand-focused) and channels drove the highest engagement and acquisition among **new customers**?
- III. **Budget Optimization:** Where should NewCo allocate its marketing spend next quarter to maximize customer growth and ROI, and what is the strategic rationale behind this shift?
- IV. **Segment Performance:** How did customer response rates vary across demographic or behavioral customer groups?

5. DATASET OVERVIEW

Data Source: Campaign_Data_Week1 Excel workbook (.xlsx) containing row-level digital interaction records.

Key Variables and Columns:

- Interaction ID (Unique interaction key)
- Campaign Type (Brand-Focused vs. Sales-Focused)
- Channel (Email, Instagram, Web banner)
- Customer Type (New vs. Existing)
- Converted (1 = Converted, 0 = Non-Converted)
- Time on Site (seconds) (Engagement duration)
- Sales (\$) (Monetary revenue generated)

6. METHODOLOGY

Following the structured BCG X Data Science Process:

- I. **Data Exploration and Cleaning:** Validate schema in Excel, audit missing values/duplicates, and format conversions/monetary metrics.
- II. **Exploratory Data Analysis (EDA):** Build pivot matrices grouping revenue, conversions, and session time by Channel and Campaign Type.
- III. **Segmentation and Driver Analysis:** Utilize Excel Slicers (filtering by Customer Type = New) to isolate drivers of new customer engagement and acquisition.
- IV. **Visualization and Synthesis:** Construct dynamic Excel visual dashboards and synthesize findings into an executive presentation.

7. TOOLS AND TECHNOLOGIES

- Data Processing and Analysis: Microsoft Excel (Pivot Tables, Formulas, Data Filtering/Slicing)
- Data Visualization and Dashboards: Microsoft Excel (Interactive Charts, Dynamic Slicers, Dashboards)
- Documentation and Presentation: Markdown, Word Documents, PowerPoint

8. ASSUMPTIONS AND CONSTRAINTS

- **Assumptions:**
 - The provided dataset accurately reflects all channel interactions and conversion activities during the campaign window.
 - Customer status (New vs. Existing) is correctly tagged within the dataset.
- **Constraints:**
 - Analysis is strictly bounded by the scope of the completed campaign data; external market trends or offline/in-store sales attributions outside the dataset are excluded unless explicitly noted.
 - Rapid turnaround expected in alignment with fast-paced BCG client engagements.